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GeoReasoner: Reasoning On Geospatially Grounded Context For Natural Language Understanding

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ABSTRACT

In human reading and communication, individuals tend to engage in geospatial reasoning, which involves recognizing geographic entities and making informed inferences about their interrelationships. To mimic such cognitive process, current methods either utilize conventional natural language understanding toolkits, or directly apply models pretrained on geo-related natural language corpora. However, these methods face two significant challenges: i) they do not generalize well to unseen geospatial scenarios, and ii) they overlook the importance of integrating geospatial context from geographical databases with linguistic information from the Internet. To handle these challenges, we propose GeoReasoner, a language model capable of reasoning on geospatially grounded natural language. Specifically, it first leverages Large Language Models (LLMs) to generate a comprehensive location description based on linguistic and geospatial information. It also encodes direction and distance information into spatial embedding via treating them as pseudo-sentences. Consequently, the model is trained on both anchor-level and neighbor-level inputs to learn geo-entity representation. Extensive experimental results demonstrate GeoReasoner's superiority in three tasks: toponym recognition, toponym linking, and geo-entity typing, compared to the state-of-the-art baselines.

CCS CONCEPTS

• **Information systems → Information retrieval**.

KEYWORDS

Natural Language Understanding, Large Language Model

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1 INTRODUCTION

Geospatial reasoning is crucial for understanding and interpreting natural language texts within the context of spatial relationships [20, 25]. For example, in the sentence "I traveled to San Jose in California to see the Tech Museum," a human can effortlessly recognize the place names "San Jose" and "California". Additionally, a human can infer that the "Tech Museum" is associated with the technology industry prevalent in the region. In practice, geospatially grounded language understanding involves essential tasks including identifying the geospatial concepts mentioned and deducing the identities of these concepts [2, 17, 22], as shown in Figure 1 (d). Therefore, such tasks heavily rely on the one-to-one correspondence between geographic entities and the physical world, especially in real-world location-based services and navigation tasks [9, 14, 28, 29, 31].

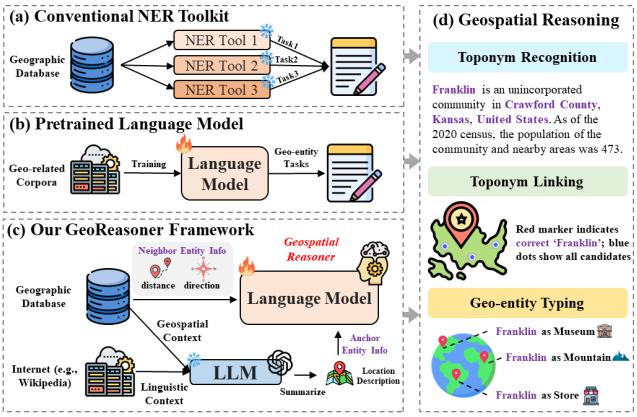

Figure 1: Comparison between our LLM-assisted GeoReasoner framework and conventional paradigms for geospatial natural language understanding task.

The rapid development of natural language understanding (NLU) has significantly impacted various fields, including geospatial reasoning [27]. NLU advancements have enabled more sophisticated approaches to understanding and interpreting natural language texts within spatial contexts, which have led to two key paradigms for tackling *geospatial reasoning tasks*, as shown in Figure 1.

a) **Conventional NLU toolkits** (e.g., Named Entity Resolution tools) are designed to identify and classify named entities in text, including geographic entities. While these tools are effective, they often struggle to generalize to unseen geospatial scenarios, which limits their applicability in dynamic and diverse contexts [23].

b) **Pretrained language models on geo-related corpora** are leveraged to understand and generate text based on extensive training data, which includes geographic information [18, 21]. However, despite their advanced capabilities, these models often overlook the importance of combining geospatial context from geographical databases with the rich linguistic information available on the Internet. This lack of integration can result in models that are less

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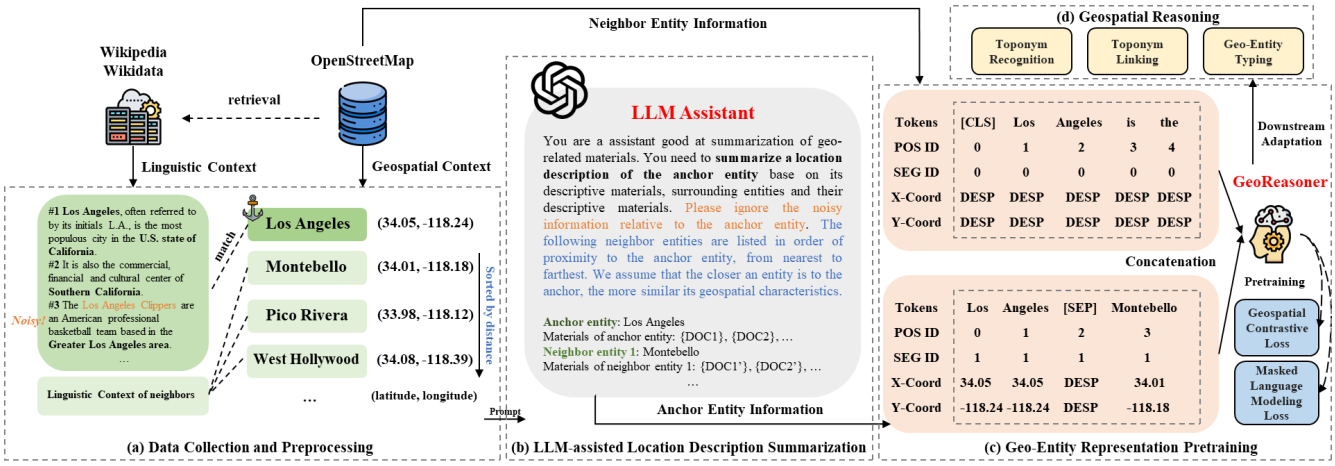

Figure 2: Overall Framework of GeoReasoner.

effective at reasoning about the spatial relationships and interdependencies between geographic entities [17].

To address the aforementioned challenges, we propose GeoReasoner, a comprehensive framework designed to improve geospatial reasoning by integrating both linguistic and geospatial contexts, as illustrated in Figure 1 (c). First, Our approach utilizes Large Language Models (LLMs) to generate detailed location descriptions by combining information from geographic databases (e.g., OpenStreetMap) and the Internet (e.g., Wikipedia). Then, GeoReasoner encodes spatial information, such as direction and distance, into spatial embeddings by treating them as pseudo-sentences. This innovative method enables the model to learn robust geo-entity representations through a dual training process that incorporates both geospatial and linguistic data. Eventually, by employing geospatial contrastive loss [24] and masked language modeling loss [4], GeoReasoner achieves superior performance in key geospatial reasoning tasks, including toponym recognition, toponym linking, and geo-entity typing, outperforming current state-of-the-art baselines. In summary, GeoReasoner represents a significant step forward by providing a robust solution for accurately understanding and reasoning on geospatial context in natural language texts.

2 METHODOLOGY

2.1 Data Collection and Preprocessing

We preprocess the training corpora by dividing them into two parts: pseudo-sentence corpora from OpenStreetMap (OSM) to provide geospatial context and natural language corpora from Wikipedia and Wikidata to provide linguistic context. As illustrated in Figure 2 (a), for the geographical dataset, we use OSM, a crowd-sourced database with extensive point geo-entities worldwide, which links to Wikipedia and Wikidata. We preprocess OSM data to gather geo-entities with these links, creating paired training data for contrastive pretraining. We linearize geospatial context by constructing pseudo-sentences for each geo-entity, sorting neighboring geo-entity names by distance. For the natural language text corpora, we use Wikipedia and Wikidata. We scrape Wikipedia articles linked from OSM annotations, break them into sentences, and use Trie-based phrase matching [13] to find sentences containing the corresponding OSM geo-entity names. Training samples are paragraphs with at least one geo-entity name. We collect Wikidata geo-entities via QID identifier and convert relation triples to natural sentences.

2.2 LLM-assisted Location Description Summarization

Due to the noisy nature of the linguistic information associated with a given entity (i.e., anchor entity), it is crucial to rewrite it by retaining the most significant parts while integrating the surrounding context. In this phase as shown in Figure 2 (b), a LLM (specifically GPT-4 Turbo in this study) is employed as a summarizer, utilizing its comprehensive capability to extract key information from both geospatial and linguistic contexts. The geospatial context includes locations and implicit geo-relations of neighboring entities to the anchor geo-entity. Meanwhile, the linguistic context encompasses essential information that specifies a geo-entity, such as sentences describing its geography, environment, culture, and history. Consequently, we define the output of the LLM as a comprehensive location description of the given anchor entity.

2.3 Geo-Entity Representation Pretraining

As illustrated in Figure 2 (c), the training process of GeoReasoner aims to learn and align anchor and neighbor entity information within the same embedding space, thereby obtaining geospatially grounded language representations. Specifically, GeoReasoner uses location descriptions as natural language inputs and generates anchor-level representations by averaging the token representations within the geo-entity name span. For neighbor entity information, GeoReasoner employs the geospatial context linearization method and spatial embedding module from SpaBERT [16], a pretrained language model that contextualizes entity representations using point geographic data. Given a anchor geo-entity and its spatial neighbors, GeoReasoner linearizes the geospatial context by sorting neighbors based on their geospatial distances from the anchor geo-entity. It then concatenates the anchor geo-entity's name with the sorted neighbors to form a pseudo-sentence. To

preserve directional relations and relative distances, GeoReasoner uses a geocoordinates embedding module, via a sinusoidal position embedding layer.

To enable GeoReasoner to process both natural language text and geographical data, we leverage a specific position embedding mechanism for each token alongside the token embeddings [17]. As shown in Figure 2 (c), *Position ID* describes the token's index position in the sentence (it starts from zero for both anchor-level and neighbor-level inputs). *Segment ID* indicates the source of the input tokens (tokens from the anchor-level input have a Segment ID of zero, while those from the neighbor-level input have a Segment ID of one). *X-coord and Y-coord* refer to inputs for the spatial coordinate embedding (tokens within the same geo-entity name span share the same X-coord and Y-coord values, but anchor-level tokens, lacking geocoordinate information, have their X-coord and Y-coord set to `DSEP`, a constant distance filler value).

During pretraining, two tasks are employed to learn geospatially grounded representations of natural language text. The first task is *geospatial contrastive learning* using an InfoNCE loss [24], which contrasts geo-entity features extracted from different contextual levels. This loss encourages GeoReasoner to generate similar representations for the same geo-entity. Consequently, each batch comprises 50% random negatives and 50% hard negatives.

Formally, Define the training data $\mathcal{D}$ consist of pairs of samples (s_i^{loc}, s_i^{geo}), where s_i^{loc} and s_i^{geo} refer to anchor-level location description and neighbor-level pseudo-sentence created from the geospatial context. Both refer to the same geo-entity. Define $f(\cdot)$ be GeoReasoner that takes both s_i^{loc} and s_i^{geo} as input and generates final representation $\mathbf{h}_i^{loc} = f(s_i^{loc})$ and $\mathbf{h}_i^{geo} = f(s_i^{geo})$. Hence, the geospatial contrastive loss $\mathcal{L}_i^{con}$ can be defined as:

$$\mathcal{L}_i^{con} = -\log \frac{e^{\text{sim}(\mathbf{h}_i^{loc}, \mathbf{h}_i^{geo})/\tau}}{\sum_{j=1}^{2N} \mathbb{1}_{[j \neq i]} e^{\text{sim}(\mathbf{h}_i^{loc}, \mathbf{h}_j^{geo})/\tau}},$$

where τ and $\text{sim}(\cdot)$ refer to temperature and cosine similarity.

Additionally, we employ a *masked language modeling* task [4] on a concatenation of the paired anchor-level sentence and neighbor-level pseudo-sentence. This task encourages GeoReasoner to recover masked tokens, thereby improving its ability to integrate and utilize information from both contexts.

2.4 Geospatial Downstream Tasks

Our study further adapts GeoReasoner to the following downstream tasks to demonstrate its ability for geospatially grounded language understanding, as shown in Figure 2 (d).

Toponym recognition involves identifying and extracting place names from unstructured text [8]. This is achieved by adding a fully connected layer to GeoReasoner and training the model on a downstream dataset, enabling it to classify each token accurately.

Toponym linking aims to identify the specific geo-entity mentioned in text by matching it to the correct record in a geographic database [8]. This database may include multiple entities with the same name as the extracted toponym from the text.

Geo-entity typing focuses on categorizing location types within a geographical database. We approach this task as a classification problem by attaching a one-layer classification head to GeoReasoner. During training, GeoReasoner predicts the type of a central geo-entity from a subregion.

3 EXPERIMENT

3.1 Experimental Setup

3.1.1 Datasets. For toponym recognition, we utilize the GeoWebNews dataset [8], which includes 200 news articles containing 2,601 toponyms with valid geocoordinates. For toponym linking, we use the Local Global Corpus [18] with 588 news articles. For geo-entity typing, we employ the dataset from [16] with geospatial context for the anchor geo-entity. The train/test ratio is 8:2.

3.1.2 Evaluation. For toponym recognition, we report precision, recall, and F1 scores at both token-level and entity-level, with entity-level accuracy requiring exact matches to the ground-truth. In toponym linking, we evaluate the accuracy of linking toponyms to their correct geo-entities in the GeoNames database, emphasizing disambiguation. For geo-entity typing, we follow [16] and report F1 score for each amenity class and overall micro F1 score.

3.1.3 Baselines. In toponym recognition, we compare GeoReasoner with fine-tuned models such as BERT [4], SimCSE-BERT [6], SpanBERT [15], SapBERT [19] and GeoLM [17]. For toponym linking, we benchmark against the same models. In geo-entity typing, we extend comparisons to include LUKE [30] and SpaBERT [16], which leverage specialized tokenization and geospatial context, respectively. All models are evaluated in their `base` versions.

3.2 Overall Performance

3.2.1 Toponym Recognition. As indicated in Table 1, GeoReasoner shows a balanced and strong performance across both token-level and entity-level metrics, especially excelling in the identification of B-topo tokens. SpanBERT excels in predicting I-topo tokens, which indicates the benefit of span prediction during pretraining for recognizing the continuation of toponyms. The results suggest that models such as GeoReasoner and GeoLM, which are specifically trained with geospatial grounding, outperform general models like BERT and its variants in toponym recognition tasks.

3.2.2 Geo-Entity Typing. The results from Table 2 clearly highlight the superior performance of the GeoReasoner and GeoLM models due to their robust performance among most types. This reveals that contrastive learning applied during pretraining has effectively facilitated the alignment of linguistic and geospatial contexts, though only geospatial context is given during inference. GeoReasoner performs worse on waste management class compared to SpanBERT, but it still exceed the performance of GeoLM, which also integrates geospatial context with linguistic context.

3.2.3 Toponym Linking. As shown in Table 3, GeoReasoner indicates the best capability in terms of top-1 retrieval (**R@1**) and top-5 retrieval (**R@5**), and the second best performance on top-10 retrieval (**R@10**). This underscores the advantage of integrating both geospatial and linguistic context in GeoReasoner compared to baseline models that emphasize one over the other. Moreover, since this task specifically necessitates models to rely solely on linguistic

GeoWebNews	Token (B-topo)			Token (I-topo)			Entity		
	Prec	**Recall**	**F1**	**Prec**	**Recall**	**F1**	**Prec**	**Recall**	**F1**
BERT	90.00	89.28	89.64	78.55	79.44	78.99	77.03	83.42	80.10
SimCSE-BERT	83.86	90.26	86.95	74.61	82.07	78.16	72.76	83.68	77.84
SpanBERT	85.98	88.37	87.16	**86.13**	**89.19**	**87.63**	75.32	81.16	78.13
SapBERT	83.12	88.32	85.64	76.26	81.11	78.61	72.48	80.16	76.12
GeoLM	91.15	90.43	90.79	79.16	84.27	81.63	82.18	85.67	83.89
GeoReasoner	**93.22**	**91.27**	**92.23**	81.56	87.90	84.61	**84.66**	**85.78**	**85.17**

Table 1: Result of toponym recognition task on GeoWebNews dataset. Bolded and underlined numbers are best and second best scores, respectively. We frame this task as a multi-class sequence tagging problem, where tokens are classified into one of three classes: *B-topo* (beginning of a toponym), *I-topo* (inside a toponym), or *O* (non-toponym).

Classes	**Edu.**	**Ent.**	**Fac.**	**Fin.**	**Hea.**	**Pub.**	**Sus.**	**Tra.**	**Was.**	**Micro F1**
BERT	67.4	63.4	**76.3**	92.9	85.6	87.2	85.6	86.2	67.8	83.5
SpanBERT	63.3	58.9	60.8	91.6	85.9	88.2	82.4	86.7	**73.5**	81.9
SimCSE-BERT	62.3	59.0	50.4	92.5	86.7	85.2	85.7	81.0	47.0	81.0
LUKE	64.8	60.8	59.8	94.5	85.7	86.7	85.4	85.1	51.7	82.5
SpaBERT	67.4	65.3	68.0	95.9	86.5	**90.0**	88.3	88.8	70.3	85.2
GeoLM	72.5	70.9	73.0	97.8	91.5	83.6	90.5	**90.8**	62.2	87.8
GeoReasoner	**73.8**	**72.2**	76.0	**98.0**	**93.3**	87.9	**92.1**	90.6	65.8	**88.9**

Table 2: Result of geo-entity typing task. Column names are the OSM classes (education, entertainment, facility, financial, healthcare, public service, sustenance, transportation and waste management). Bolded and underlined numbers are best and second best scores.

LGL	**R@1**	**R@5**	**R@10**
BERT	34.6	67.5	**78.1**
RoBERTa	24.2	48.7	60.6
SpanBERT	25.2	48.8	61.0
SapBERT	30.8	58.8	72.2
GeoLM	38.2	65.3	72.6
GeoReasoner	**40.1**	**68.3**	75.9

Table 3: Result of toponym linking task on LGL dataset. Bolded and underlined numbers are best and second best scores, respectively. R@k measures whether ground-truth GeoNames ID presents among the top k retrieval results.

Ablation Settings	**Prec**	**Recall**	**F1**
GeoReasoner	84.66	85.78	85.17
GeoReasoner w/o contrastive loss	73.67	79.45	76.45
GeoReasoner w/o MLM loss	78.23	80.22	79.21
GeoReasoner w/o spatial embed.	76.98	85.31	80.93
GeoReasoner w/o LLM summarization	79.12	83.42	81.17

Table 4: Ablation study on toponym recognition task.

context, GeoReasoner's geospatial contrastive learning paradigm proves beneficial by facilitating alignment among sources.

3.3 Ablation Study

Four ablation experiments are conducted on toponym recognition task: i) removing the geospatial contrastive loss; ii) removing the masked language modeling loss; iii) removing the spatial coordinate embedding layer that leverages geo coordinates as input; iv) removing the inclusion of LLM for location description summarization (i.e., linguistic context will be directly used as anchor-level input for GeoReasoner). Results from Table 4 demonstrate a significant drop in performance when either the contrastive loss or the masked language modeling loss is removed. This underscores their crucial roles during pretraining. Furthermore, all metrics decrease when the spatial coordinate embedding layer is omitted, indicating that geocoordinates encapsulate implicit geospatial context. Additionally, performance declines without LLM-based summarization, likely due to the noisy nature of the original linguistic context.

4 RELATED WORK

Research on understanding geospatial concepts in natural language has used general-purpose NER tools like Stanford NER [5] and NeuroNER [3], as well as geospatial-specific tools such as the Edinburgh Geoparser [10, 26] and Mordecai [11, 12], to identify and link toponyms to geographical databases. Deep learning models, including TopoCluster [1] and CamCoder [7], have connected toponyms with geographic locations using geodesic and lexical features. However, these models often lack effective generalizability of unseen geospatial context during inference. Furthermore, SpaBERT [16] serves as a language model trained on geographical corpora, but it cannot leverage linguistic information from the Internet to obtain comprehensive geo-entity representation. GeoLM [17] manages to fuse geospatial and linguistic information together via projecting them into a joint embedding. Our GeoReasoner framework stands out by harnessing LLMs' powerful summarization abilities to derive anchor entity information as linguistic context, while seamlessly incorporating geospatial context through concatenation.

5 CONCLUSION

GeoReasoner effectively addresses the challenges of geospatial reasoning in natural language processing by integrating linguistic information with geospatial context. It leverages powerful LLMs to generate location descriptions and encodes spatial information as pseudo-sentences. In future research, we will focus on integrating geospatial reasoning with the reasoning capabilities of LLMs.

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